File Handling in C

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Introduction

What is a File?

- A named collection of data, typically stored in a secondary storage (e.g., hard disk).

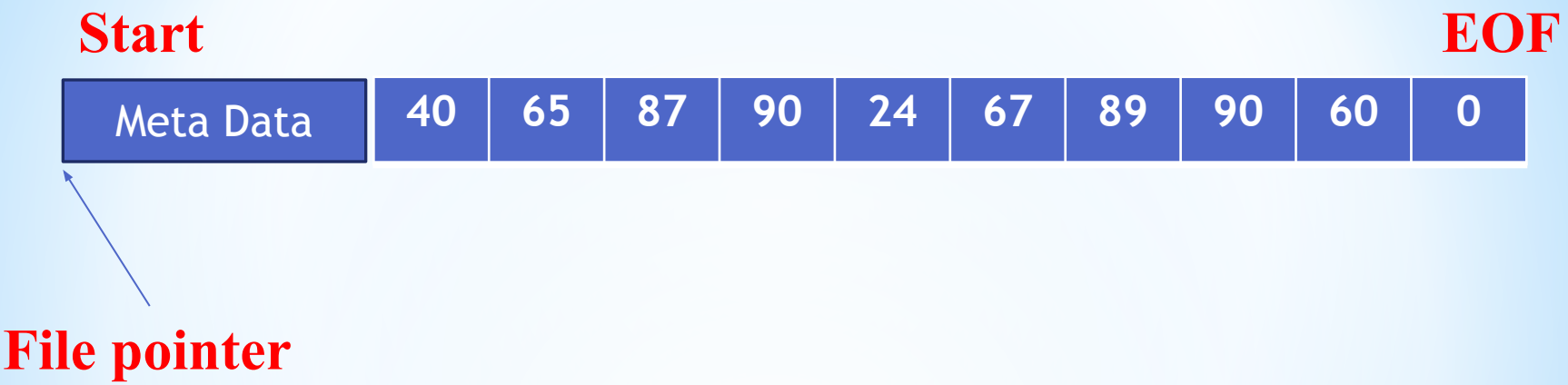
  Examples
  - Records of all employees in an organization
  - Document files created using Microsoft Word
  - Video of a movie
  - Audio of a music

- Non-volatile data storage
  - Can be used when power to computer is off

How a File is Stored?

- Stored as sequence of bytes, logically contiguous (may not be physically contiguous on disk).
  - Discrete storage unit for data in the form of a stream of bytes.
  - Every file is characterized with a starting of file, sequence of bytes (actual data), and end of stream (or end of file).
  - Allow only sequential access of data by a pointer performing.
  - Meta-data (information about the file) before the stream of actual data can be maintained to have a knowledge about the data stored in it.

How a File is Stored?



Note:

- The last byte of a file contains the end end-of-file character (EOF, with ASCII code 1A (Hex).
- While reading a file, the EOF character can be checked to know the end.

Type of Files

- Text files
  - Contain ASCII code only
    - C-programs
- Binary files
  - Contain non-ASCII characters
    - Image, audio, video, executable, etc.

File Handling in C

Operations on Files

- Typical operations on a file are
  - Open : To open a file to store/retrieve data in it
  - Read : The file is used as an input
  - Write : The file is used as output
  - Close : Preserve the file for a later use
  - Access: Random accessing data in a file

Opening and Closing a File

File Handling Commands

- Include header file `<stdio.h>` to access all file handling utilities.
- A data type namely FILE is there to create a pointer to a file.

Syntax

```
FILE * fptr;        // fptr is a pointer to file
```

- To open a file, use `fopen()` function

Syntax

```
FILE * fopen(char *filename, char *mode)
```

- To close a file, use `fclose()` function

Syntax

```
int fclose(FILE *fptr);
```

`fopen()` function

- The first argument is a string to characters indicating the name of the file to be opened.
  - The convention of file name should follow the convention of giving file name in the operating system.

Examples:

```
xyz12.c   student.dataFile PDS.txt

myFile
```

`fopen()` function

- The second argument is to specify the mode of file opening. There are five file opening modes in C
  - "r" : Opens a file for reading
  - "w" : Creates a file for writing (overwrite, if it contains data)
  - "a" : Opens a file for appending - writing on the end of the file
  - “rb” : Read a binary file (read as bytes)
  - “wb” : Write into a binary file (overwrite, if it contains data)
- It returns the special value NULL to indicate that it couldn't open the file.

`fopen()` function

- If a file that does not exist is opened for writing or appending, it is created as a new.
- Opening an existing file for writing causes the old contents to be discarded.
- Opening an existing file for appending preserves the old contents, and new contents will be added at the end.
- File opening error
  - Trying to read a file that does not exist.
  - Trying to read a file that doesn't have permission.
  - If there is an error, fopen() returns NULL.

Example: fopen()

```
#include <stdio.h>
void main()
{
   FILE *fptr;                    // Declare a pointer to a file

   char filename[]= "file2.dat";

   fptr = fopen(filename,"w");

// Also, alternatively
//     fptr = fopen ("file2.dat","w");

   if (fptr == NULL) {
      printf ("Error in creating file");
      exit(-1);                   // Quit the function
   }
   else /* code for doing something */
}
```

Reading from a File

Reading from a File

- Following functions in C (defined in `stdio.h`) are usually used for reading simple data from a file
  - `fgetc(…)`
  - `fscanf(…)`
  - `fgets(…)`
  - `getc(…)`
  - `ungetc(…)`

Reading from a File: `fgetc()`

Syntax for `fgetc(…)`

```
int fgetc(FILE *fptr)
```

- The `fgetc()` function returns the next character in the stream `fptr` as an `unsigned char` (converted to `int`).
- It returns `EOF` if end of file or error occurs.

```
FILE *fptr;
int c;
/* Open file and check it is open */
while ((c = fgetc(fptr)) != EOF)
   {
     printf ("%c",c);
   }
```

Reading from a File: `fscanf()`

Syntax for `fscanf(...)`

```
int fscanf(FILE *fptr, char *format, ...);
```

- `fscanf` reads from the stream `fptr` under control of format and assigns converted values through subsequent assignments, each of which must be a pointer.
  - It returns when format is exhausted.
- `fscanf` returns `EOF` if end of file or an error occurs before any conversion.
- it returns the number of input items converted and assigned.

Example: Using `fscanf(...)`

```
FILE *fptr;

fptr= fopen ("input.dat","r");
int n;
/* Check it's open */
if (fptr == NULL)
   {
    printf("Error in opening file \n");
   }


  n = fscanf(fptr,"%d %d",&x,&y);

  ...
```

input.dat

20 30 40 50

x = 20
x = 30

Reading from a File: `fgets(...)`

Syntax for `fgets(...)`

```
char *fgets(char *s, int n, FILE *fptr)
```

- `s` The array where the characters that are read will be stored.
- `n` The size of *s*.
- `fptr` The stream to read.

- `fgets()` reads at most `n-1` characters into the array `s`, stopping if a newline is encountered.
  - The newline is included in the array, which is terminated by '\0'.
- The `fgets()` function returns `s` or NULL if EOF or error occurs.

Example: Using `fgets(...)`

```
FILE *fptr;
char line [1000];
/* Open file and check it is open */

while (fgets(line,1000,fptr) != NULL)
   {
   printf ("Read line %s\n",line);
   }
```

Reading a File: `getc(…)`

Syntax for `getc(…)`

```
int getc(FILE *fptr)
```

- `getc(…)` is equivalent to `fgetc(…)` except that it is a macro.

Example: Using `getc(...)`

C program to read a text file and then print the content on the screen.

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
   int ch, fileName[25];
   FILE *fp;
   printf("Enter the name of file you wish to read\n");
   gets(fileName);
   fp = fopen(fileName,"r");  // read mode

  if( fp == NULL )
   {
      printf("Error while opening the file.\n");
      exit(-1);
   }

   printf("The contents of %s f
   while( ( ch = getc(fp) ) !=
      printf("%c",ch);

   fclose(fp);
   return 0;
}
```

OUTPUT

```
Enter the name of file you wish to read
test.txt
The contents of test.txt file are :
C programming is fun.
```

Undo a File Reading: `ungetc()`

`ungetc()`: Push a character back onto an input stream.

Syntax:

```
int ungetc(int c, FILE *fptr)
```

Arguments:

- `c` The character that you want to push back.
- `fptr` The stream you want to push the character back on.

- Only one character of pushback is guaranteed per file.
- `ungetc` may be used with any of the input functions like `scanf`, `getc`, or `getchar`.

Example: `ungetc()`

```
#include <stdio.h>
int main(void)
{
    int ch;

    while ((ch = getchar()) != '1') // reads characters from the stdin
        putchar(ch);                // and show them on stdout until enc

    ungetc(ch, stdin); // ungetc() returns '1' previously read back to stdin

    ch = getchar(); // getchar() attempts to read next character from stdin
                    // and reads character '1' returned back to the stdin by unge

    putchar(ch);    // putchar() displays character
    puts("");

    printf("Thank you!\n");
    return 0;
}
```

OUTPUT

```
a
a
v
v
c
c
u
u
1
1
Thank you!
```

Writing into a File

Writing into a File

- Following functions in C (defined in `stdio.h`) are usually used for writing simple data into a file
  - `fputc(…)`
  - `fprintf(…)`
  - `fputs(…)`
  - `putc(…)`

Writing into a File: `fputc(...)`

Syntax for `fputc(...)`

```
int fputc(int c, FILE *fptr)
```

- The `fputc()` function writes the character `c` to file `fptr` and returns the character written, or EOF if an error occurs.

```
#include  <stdio.h>

   filecopy(File *fpIn, FILE *fpOut)
   {
       int c;
       while ((c = fgetc(fpIn) != EOF)
           fputc(c, fpOut);
   }
```

Writing into a File: `fprintf(…)`

Syntax for `fprintf(…)`

```
int fprintf(FILE *fptr, char *format,...)
```

- `fprintf()` converts and writes output to the steam `fptr` under the control of `format`.
- The function is similar to `printf()` function except the first argument which is a file pointer that specifies the file to be written.
- The `fprintf()` returns the number of characters written, or negative if an error occur.

Writing into a File: `fprintf(…)`

```
#include  <stdio.h>

   void main()
   {
      FILE *fptr;
      fptr = fopen("test.txt", "w");

      fprintf(fptr, "Programming in C is really a fun!\n");
      fprintf(fptr, "Let's enjoy it\n");

      fclose(fptr);

      return;
   }
```

Writing into a File: `fputs()`

Syntax for `fputs`:

```
int fputs(char *s, FILE *fptr)
```

- The `fputs()` function writes a string (which need not contain a newline) to a file.
- It returns non-negative, or EOF if an error occurs.

Example: fputs(...)

```
#include  <stdio.h>

   void main()
   {
      FILE *fptr;
      fptr = fopen("test.txt", "w");

      fputs("Programming in C is really a fun!", fptr);
      fputs("\n", fptr);
      fputs("Let's enjoy it \n", fptr);

      fclose(fptr);

      return;
   }
```

Writing into a File: `putc(...)`

Syntax for `putc(...)`

```
int putc(FILE *fptr)
```

- The `putc()` function is same as the `putc(...)`.

```
#include  <stdio.h>

   filecopy(File *fpIn, FILE *fpOut)
   {
       int c;
       while ((c = getc(fpIn) != EOF)
           putc(c, fpOut);
   }
```

Writing into a File: Example

- A sample C program to write some text reading from the keyboard and writing them into a file and then print the content from the file on the screen.

```
#include  <stdio.h>

   main()
   {
       FILE *f1;
       char c;
       printf("Data Input\n\n");
         /* Open the file INPUT */

       f1 = fopen("INPUT", "w");
```

Contd…

Writing into a File

```
while((c=getchar()) != EOF)  /* Get a character from keyboard*/
putc(c,f1);  /* Write a character to IN
fclose(f1);  /* Close the file INPUT*/
printf("\nData Output\n\n")
f1 = fopen("INPUT","r");  /* R
while((c=getc(f1)) != EOF)
printf("%c",c);  /* Display a chara
fclose(f1);  /* Close the file INPUT */
}
```

OUTPUT

```
Data Input
   This is a program to test
the file handling features on
this system

Data Output
   This is a program to test
the file handling features on
this system
```

Special Streams in C

Special Streams

- When a C program is started, the operating system environment is responsible for opening three files and providing file pointer for them. These files are

  - `stdin` Standard input. Normally it is connected to keyboard
  - `stdout` Standard output, In general, it is connected to display screen
  - `stderr` It is also an output stream and usually assigned to a program in the same way that `stdin` and `stderr` are. Output written on `stderr` normally appears on the screen

Note:

`getc(stdin)` is same as `fgetc (stdin)`

Special Streams

```
fprintf (stdout,"Hello World!\n");

printf(“"Hello World!\n");
```

The above two statements are same!

Example: Special Streams

```
#include <stdio.h>
main()
{
 int i;

 fprintf(stdout,"Give value of i \n");
 fscanf(stdin,"%d",&i);
 fprintf(stdout,"Value of i=%d \n",i);

 }
```

OUTPUT

Give value of i
15
Value of i=15

Error Handling : `stderr` and `exit`

- What happens if the errors are not shown in the screen instead if it's going into a file or into another program via a pipeline.
- To handle this situation better, a second output stream, called `stderr`, is assigned to a program in the same way that `stdin` and `stdout` are.
- Output written on `stderr` normally appears on the screen even if the standard output is redirected.

Example: Error Handling

```
#include <stdio.h>

   /* cat:  concatenate files */
   main(int argc, char *argv[])
   {
       FILE *fp;
       void filecopy(FILE *, FILE *);
       char *prog = argv[0];  /* program name for errors */

       if (argc == 1 ) /* no args; copy standard input */
           filecopy(stdin, stdout);
       else
           while (--argc > 0)
```

Contd…

Example: Error Handling

```
  if ((fp = fopen(*++argv, "r")) == NULL) {
      fprintf(stderr, "%s: can't open %s\n", prog, *argv);
      exit(1);
  } else {
        filecopy(fp, stdout);
        fclose(fp);
       }

 if (ferror(stdout)) {
       fprintf(stderr, "%s: error writing stdout\n", prog);
       exit(2);
  }
       exit(0);
}
```

Direct Input and Output

Structured Input/Output for Files

- Other than the simple data, C language provides the following two functions for storing and retrieving composite data.

- `fwrite()` To write a group of structured data

- `fread()` To read a group of structured data

Writing Records: `fwrite()`

`fwrite()` writes data from the array pointed to, by `ptr` to the given stream `fptr`.

Syntax:

```
int fwrite(void *ptr, int size, int nobj, FILE *fptr);
```

- `ptr` This is the pointer to a block of memory with a minimum size of `size *nobj` bytes.
- `size` This is the size in bytes of each element to be written.
- `nobj` This is the number of elements, each one with a size of `size` bytes.
- `fptr` This is the pointer to a FILE object that specifies an output stream.

Example: fwrite()

```
#include<stdio.h>

      struct Student
      {
             int roll;
             char name[25];
             float marks;
      };

      void main()
      {
             FILE *fp;
             int ch;
             struct Student Stu;

             fp = fopen("Student.dat","w");    //Statement   1

             if(fp == NULL)
             {
                    printf("\nCan't open file or file doesn't exist ");
                    exit(0);
             }
```

Contd…

Example: fwrite()

```
do
        {
                printf("\nEnter Roll : ");
                scanf("%d",&Stu.roll);

                printf("Enter Name : ");
                scanf("%s",Stu.name);

                printf("Enter Marks : ");
                scanf("%f",&Stu.marks);

                fwrite(&Stu,sizeof(Stu),1,fp);

                printf("\nDo you want to add another data (y/n) : ");
                ch = getchar();

        }while(ch=='y' || ch=='Y');

        printf("\nData written successfully...");

        fclose(fp);
    }
```

Contd…

Example: `fwrite()`

OUTPUT

```
Enter Roll : 1
Enter Name : AA
Enter Marks : 78.53
Do you want to add another data (y/n) : y

Enter Roll : 2
Enter Name : BB
Enter Marks : 72.65
Do you want to add another data (y/n) : y

Enter Roll : 3
Enter Name : CC
Enter Marks : 82.65
Do you want to add another data (y/n) : n

Data written successfully...
```

Reading Records: `fread()`

`fread()` reads data from the given stream into the array pointed to, by `ptr`.

Syntax:

```
int fread(void *ptr, int size, int nobj, FILE *fptr);
```

- `ptr` This is the pointer to a block of memory with a minimum size of `size *nobj` bytes.
- `size` This is the size in bytes of each element to be read.
- `nobj` This is the number of elements, each one with a size of `size` bytes.
- `fptr` This is the pointer to a FILE object that specifies an input stream.

Example: fread()

```
#include<stdio.h>

        struct Student
        {
                int roll;
                char name[25];
                float marks;
        };

        void main()
        {
                FILE *fp;
                int ch;
                struct Student Stu;

                fp = fopen("Student.dat","r");   //Statement   1

                if(fp == NULL)
                {
                  printf("\nCan't open file or file doesn't exist.");
                  exit(0);
                }
```

Contd…

Example: fread()

```
        printf("\n\tRoll\tName\tMarks\n");

        while(fread(&Stu,sizeof(Stu),1,fp)>0)

printf("\n\t%d\t%s\t%f",Stu.roll,Stu.name,Stu.marks);

       fclose(fp);
       }
```

OUTPUT

Roll	Name	Marks
1	AA	78.53
2	BB	72.65
3	CC	82.65

Random Accessing Files

File Positioning Functions in C

- When doing reads and writes to a file, the OS keeps track of where you are in the file using a counter generically known as the file pointer.

- So long we have learnt about the sequential access in a file.
- The following are the functions to access file at random
- `ftell()` Tell the current position of the file pointer
- `fseek()` To position a file pointer at a desired place within the file
- `rewind()` Is equivalent to `fseek()`

Random Accessing a File: `ftell()`

```
long ftell(FILE *fptr);
```

- `ftell()` takes a file pointer `fptr` and returns in a number of type `long`, that corresponds to the current position.
- It returns `-1L` on error.

Example

```
long n;
n = ftell(fptr);
```

Note:

In this case, `n` gives the relative offset (in bytes) of the current position. This means that `n` bytes have already been read (or written).

Random Accessing a File: `fseek()`

```
int fseek(FILE *fptr, long offset, int whence);
```

- `fseek()` function is used to move the file position to a desired location within the file.
- The first argument is the file in question. `offset` argument is the position that you want to seek to, and `whence` is what that offset is relative to.
- You can set the value of `whence` to one of the three things:

SEEK_SET	offset is relative to the beginning of the file.
SEEK_CUR	offset is relative to the current file pointer position.
SEEK_END	offset is relative to the end of the file.

Example: `fseek()`

- You can set the value of `whence` to one of the three things:

```
fseek(fp, 0L, SEEK_SET);    // go to the beginning
fseek(fp, 0L, SEEK_CUR);    // Stay at the current position
fseek(fp, 0L, SEEK_END);    // go to the end of the file, i.e., past
                               the last character of the file
fseek(fp, 0L, SEEK_SET);    // go to the beginning

fseek(fp, m, SEEK_SET);    // Move to (m+1)th  byte in the file
fseek(fp, m, SEEK_CUR);    // Go forward by m bytes
fseek(fp, -m, SEEK_CUR);    // Go backward by m bytes from the
                               current position
fseek(fp, -m, SEEK_END);    // Go back by m bytes from the end
```

Random Accessing a File: `rewind()`

```
void rewind(FILE *fptr);
```

- `rewind()`: It repositions the file pointer at the beginning of the file

Example

```
rewind (fptr);                  // Set the file pointer at the beginning
fseek(fptr, 0L, SEEK_SET);      // same as the rewind()
```

fseek() vs. rewind()

Return value

- For `fseek()`, on success zero is returned; `-1L` is returned on failure.
- The call to `rewind()` never fails.

Examples:

```
fseek(fp, 100, SEEK_SET); // seek to the 100th byte of the file
fseek(fp, -30, SEEK_CUR); // seek backward 30 bytes from the current position
fseek(fp, -10, SEEK_END); // seek to the 10th byte before the end of file

fseek(fp, 0, SEEK_SET);   // seek to the beginning of the file
rewind(fp);               // seek to the beginning of the file
```

Examples

File Handling : Example 1

A program to copy a text file to another file.

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
   char ch, sourceFile[20], targetFile[20];
   FILE *source, *target;

   printf("Enter name of file to copy\n");
   gets(sourceFile);

   source = fopen(sourceFile, "r");

   if( source == NULL )
   {
      printf("Input file error. Program abort...\n");
      exit(-1);
   }
```


Contd…

File Handling : Example 1

```
    printf("Enter name of target file\n");
    gets(target_file);

    target = fopen(targetFile, "w");

    if( target == NULL )
    {
       fclose(source);
       printf("Output File Error! File copy fails...\n");
       exit(-1);
    }

    while( (ch = fgetc(source) ) != EOF )
       fputc(ch, target);

    printf("File copied successfully.\n");

    fclose(source);
    fclose(target);

    return 0;
}
```

File Handling : Example 2

A program to copy a text file to another file. Read the file names through command line.

```
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char *argv[])
{
   char ch;
   FILE *source, *target;

   source = fopen(argv[1], "r");

   if( source == NULL )
   {
      printf("Input file error. Program abort...\n");
      exit(-1);
   }
```


Contd…

File Handling : Example 2

```
    target = fopen(argv[2], "w");

    if( target == NULL )
    {
        fclose(source);
        printf("Output File Error! File copy fails...\n");
        exit(-1);
    }

    while( (ch = fgetc(source) ) != EOF )
        fputc(ch, target);

    printf("File copied successfully.\n");

    fclose(source);
    fclose(target);

    return 0;
}
```

File Handling : Example 3

A program to concatenate a file (say A) to another file (say B) so that the resultant file is A = A+B. Read the file names for A and B through command line.

```
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char *argv[])
{
   int ch;
   FILE *fpA, *fpB;

   fpB = fopen(argv[2], "r");    //Open the file B

   if( fpB == NULL )
   {
      printf("Input file error. Program abort...\n");
      exit(-1);
   }
```


Contd…

File Handling : Example 3

```
    fpA = fopen(argv[1], "a");  //Open the file A in append mode

    if( fpA == NULL )
    {
       fclose(fpA);
       printf("Output File Error! File merging fails...\n");
       exit(-1);
    }

    while( (ch = fgetc(fpA) ) != EOF )
       fputc(ch, fpA);

    printf("Files are concatenated successfully.\n");

    fclose(fpA);
    fclose(fpB);

    return 0;
}
```

File Handling : Example 4

A program to encrypt a text file. Read the file names through command line.

```
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char *argv[])
{
   char ch;
   FILE *source, *target;

   source = fopen(argv[1], "r");

   if( source == NULL )
   {
      printf("Input file error. Program abort...\n");
      exit(-1);
   }
```


Contd…

File Handling : Example 4

```
    target = fopen(argv[2], "w");

    if( target == NULL )
    {
        fclose(source);
        printf("Output File Error! File copy fails...\n");
        exit(-1);
    }

    while( (ch = fgetc(source) ) != EOF )
        fputc(ch+10, target);      //Change the character...

    printf("File copied successfully.\n");

    fclose(source);
    fclose(target);

    return 0;
}
```

File Handling : Example 5

A program to display a file on the screen. Read the file name through command line.

```
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char *argv[])
{
   char ch;
   FILE *source, *target;

   source = fopen(argv[1], "r");

   if( source == NULL )
   {
      printf("Input file error. Program abort...\n");
      exit(-1);
   }

   while( (ch = fgetc(source) ) != EOF )
      fputc(ch);

   fclose(source);

   return 0;
}
```

File Handling : Example 6

A program to store a record in file. Read the file and store all records in an array.

```
#include <stdio.h>
#include <stdlib.h>

struct Student {
     int rollNo;
     char name[20];
     float marks;
};

int main(int argc, char *argv[])
{
   int choice = 1;
   struct Student *data;
   FILE *outfile, *infile;

   outfile = fopen(argv[1], "w");

   if( outfile == NULL )
   {
      printf("Input file error. Program abort...\n");
      exit(-1);
   }
```


Contd…

File Handling : Example 6

A program to store a record in file. Read the file and store all records in an array.

```
   while (choice) {
       data = (struct *)malloc(sizeof(struct Student));
       if (data != NULL) {
          printf(\nEnter Roll No: "); scanf("%d",&data->rollNo;);
          printf(\nEnter Name: "); scanf("%s",data->name;);

          fwrite (data, sizeof(struct Student), 1, outfile);

          printf("\nDo you want to add more record (Type 0 for NO)?");
          scanf("%d", &choice);
       }
    }

    fclose(outfile);


   return 0;
}
```


Contd…

File Handling : Example 6

```
    infile = fopen(argv[1], "r");

    struct Student data[100];

    if( infile == NULL )
    {
      printf("File error. Program abort...\n");
      exit(-1);
    }
     choice = 0;

     while (fread (&data, sizeof(struct Student), 1, infile))
         data[choice++] = data;
    }

   return 0;
}
```

Any question?

You may post your question(s) at the “Discussion Forum” maintained in the course Web page.

Problems to Ponder…

Problems for Practice…